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Linear Regression

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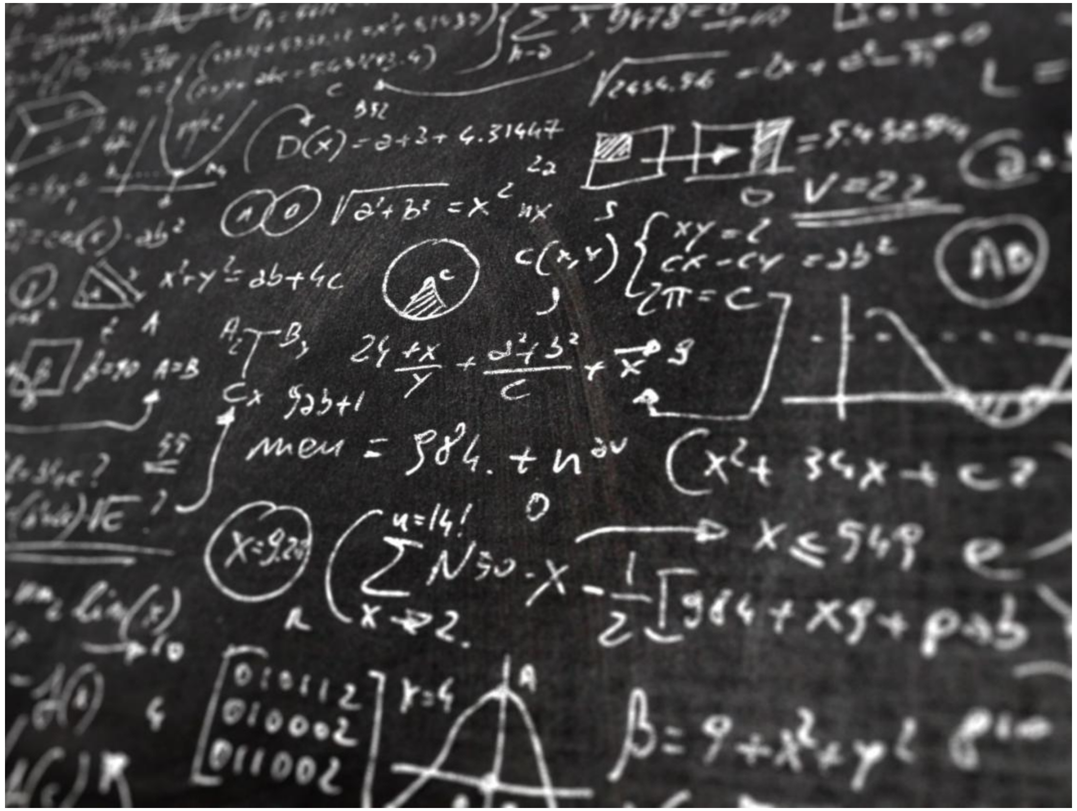
What is a model?

Models are typically thought of as an **abstraction** of (or an approximation to) **reality**.

Models allow us to separate data into predictable and unpredictable elements.

By **learning from patterns** that occurred in the **past**, we are often able to **predict the future**.



Why do we need a model?

Help **understand the past**, **learn** from it, and then **predict the future**

Many business **decisions** are **driven** by our **ability** to **anticipate the future** and customer’s behavior or competitors’ moves.

For example;

“How many customers will visit my store tomorrow?”

“Should I manufacture 1,000 or 10,000 smart phones?”

“If I invest my money in stock A today, will its value go up tomorrow?”

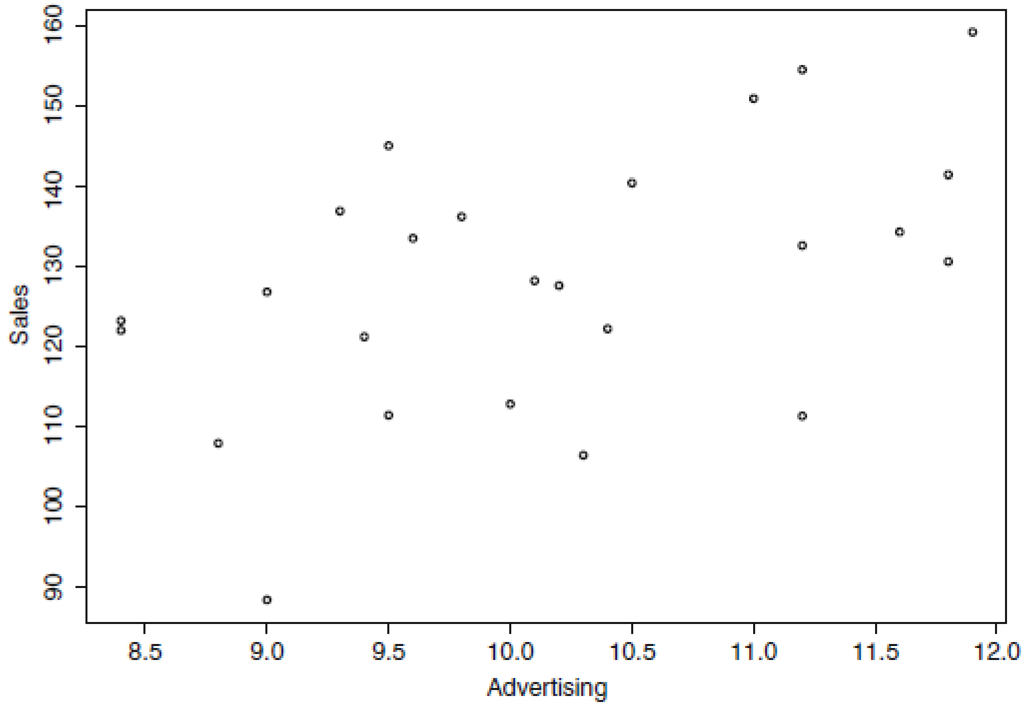
Example

- For example:
  - information on sales of a particular type of soft drink distributed across different sales regions.
  - In addition to sales information, the table also records the amount of money spent on advertising in each region.
  - Are your advertising efforts effective?
  - Is there a positive relationship between advertising and sales? (i.e., the more money spent on advertising, the more sales grow).

Sales	Advt
145.1	9.5
128.3	10.1
121.3	9.4
134.4	11.6
106.5	10.3
111.5	9.5
132.7	11.2
126.9	9
151	11

Example

- Is there a positive relationship between advertising and sales?
  - (i.e., the more money spent on advertising, the more sales grow).
- For every dollar we spend on advertising, do we get enough in return (via sales)?
- How can we quantify the impact of advertising?



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Model?

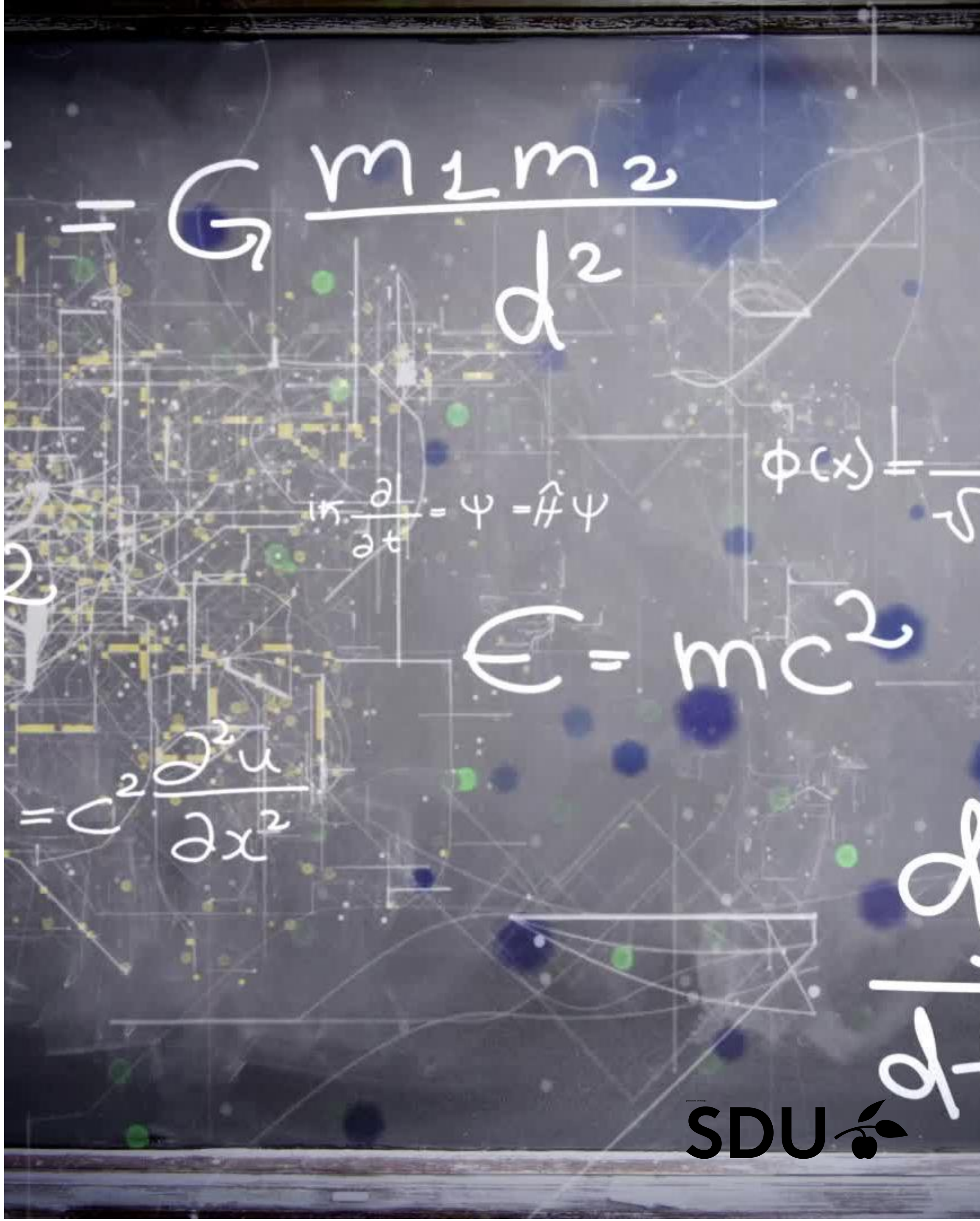
A model – in its simplest form – is a mathematical equation.

For example, a (hypothetical) model could look like the following equation:

Sales = $10,000+5×Advertising

The quality of this prediction depends on the quality of the model: better models will result in predictions that are closer to the truth!

– poor data on the input side will result in poor



Model Coefficients

For an equation:

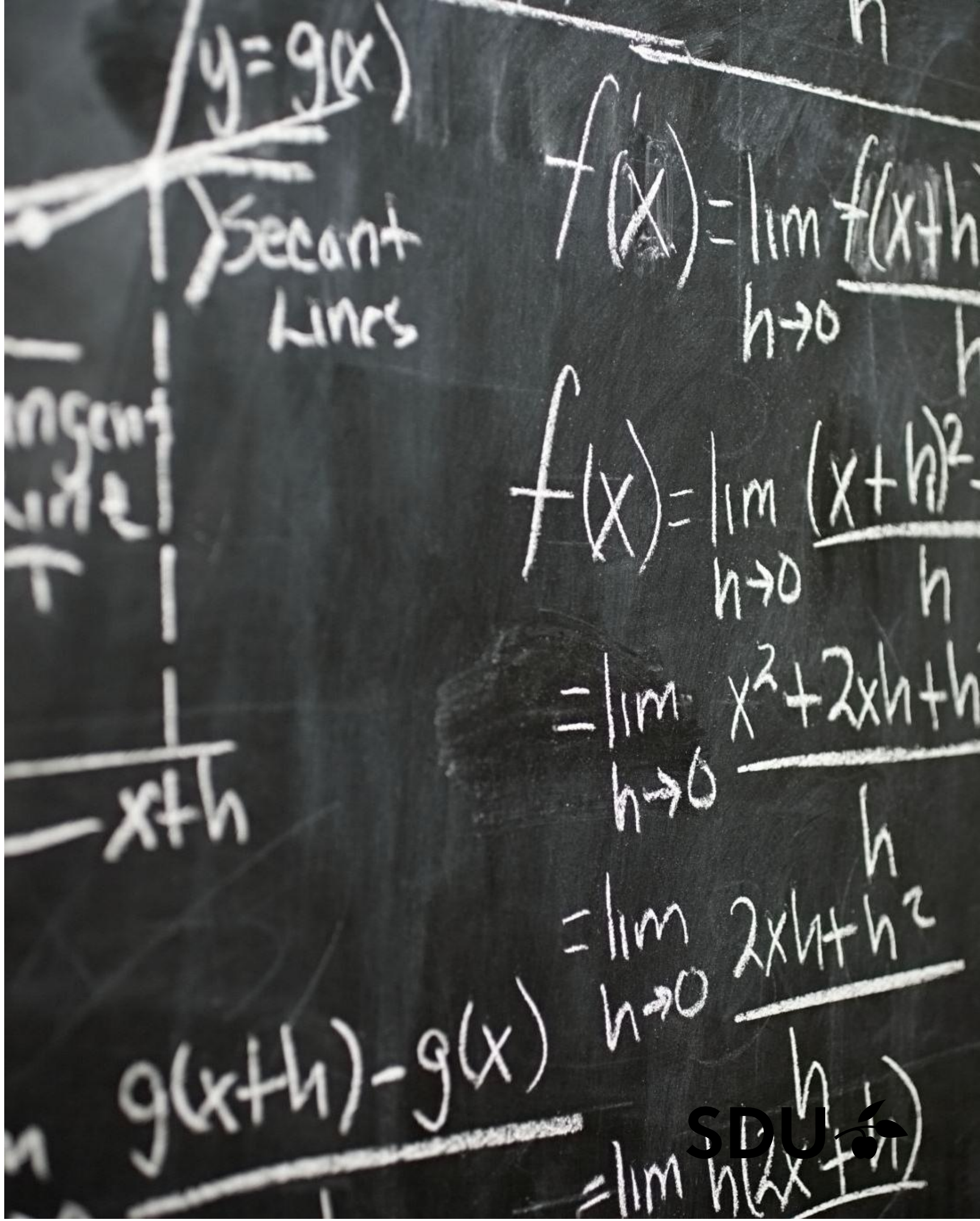
Sales = \$10,000+5×Advertising

What does the number “\$10,000” tells us?

the value \$10,000 is also referred to as the intercept, indicates where the model “**intercepts**” the Y-axis (i.e., the value where the response equals zero).

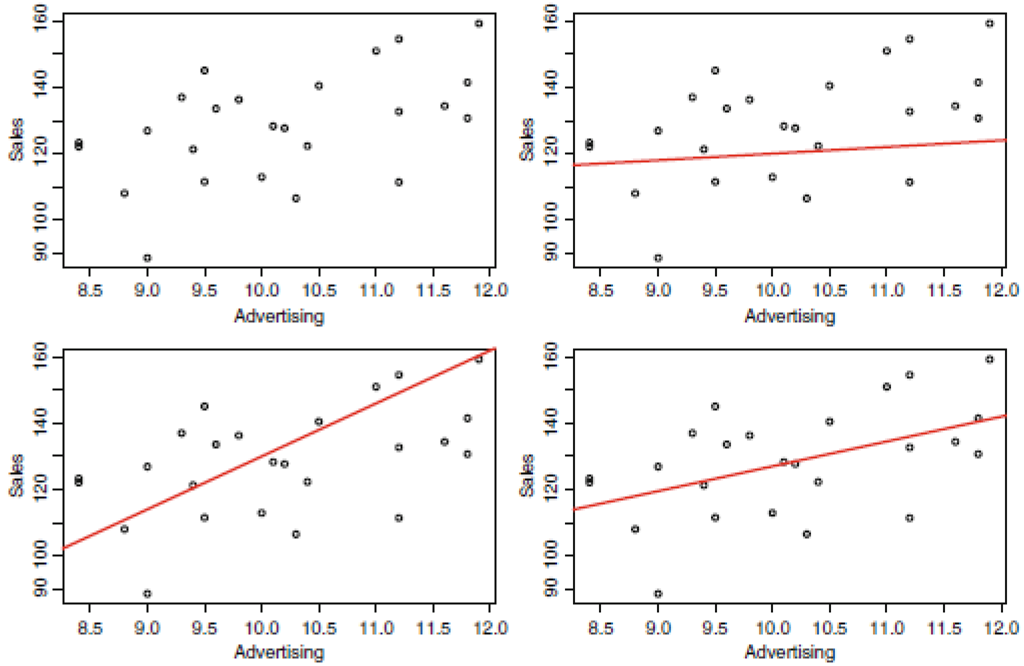
What does the number “5” tells us?

It measures how fast sales increase for each additional dollar we spend on advertising. It is called as the **slope of the model**

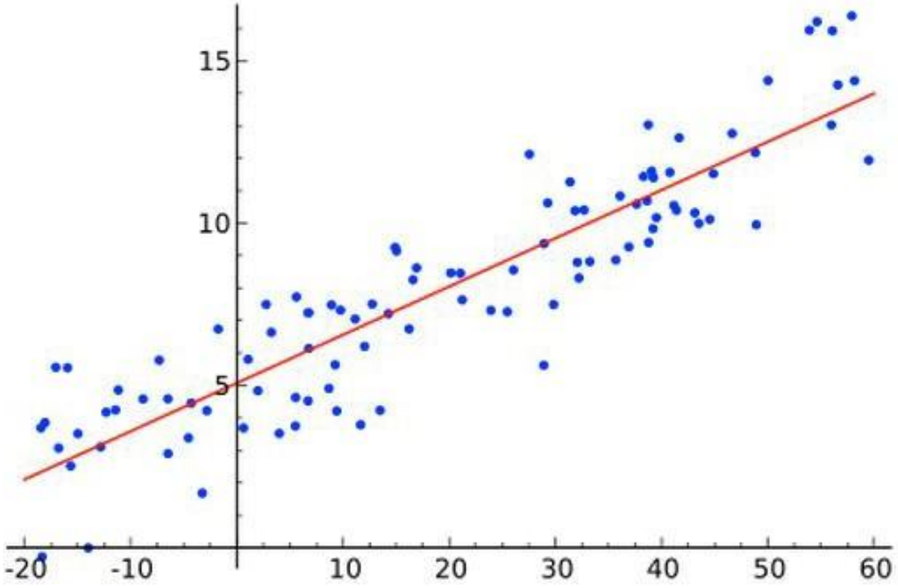


Fitting and Interpreting a Regression Model

- Usually difficult to understand the results of models and to use them for decision making.
- Three potential models for that data. Which one is the best?
- Quantify how well that line fits the data?



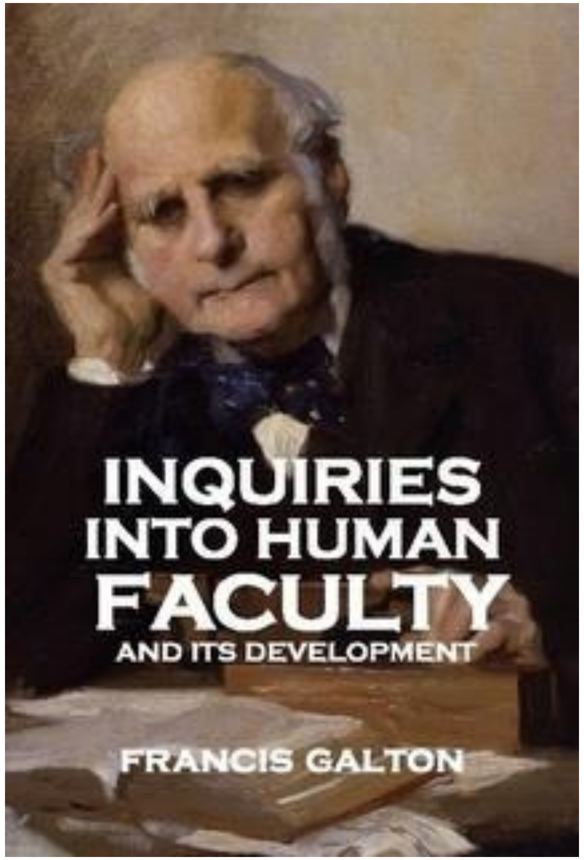
Regression ~ *predictive modeling*



Regression analysis

From Wikipedia, the free encyclopedia

In statistics, **regression analysis** is a statistical technique for estimating the relationships among variables. It includes many techniques for modeling and analyzing several variables, when the focus is on the relationship between a dependent variable and one or more independent variables. More specifically,



making an assumption of linear dependence on the inputs

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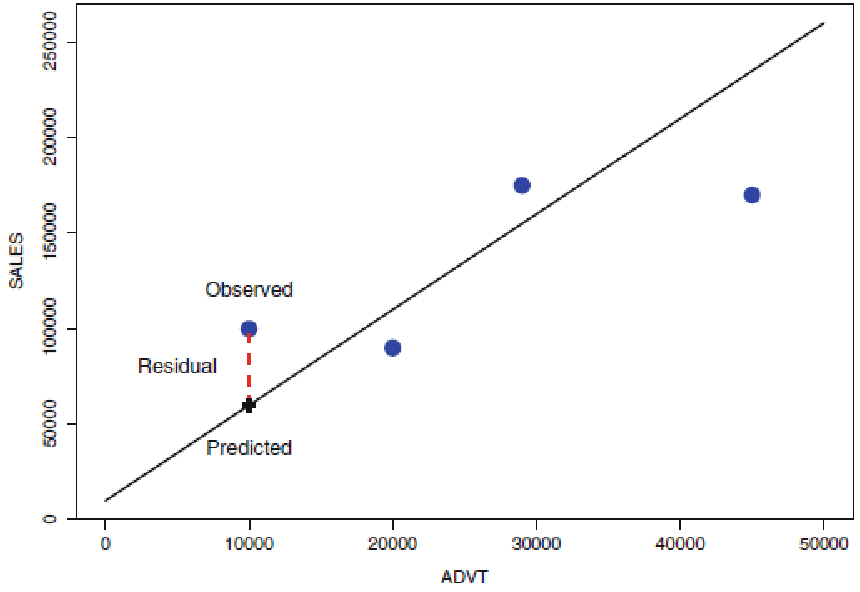
Least Squares Regression

- The difference between the **observed** and the **predicted** values measures how well our model fits the data; smaller differences imply a better fit.
    - The difference between observed and predicted points is referred to as the **residual**.

- Least squares regression determines the **best model** by finding **the line** that *minimizes the sum of all squared residuals*

- It is important to realize that this is just one way of defining "the best" model;
    - another way would be to minimize the sum of the absolute residuals, and there are even other criteria



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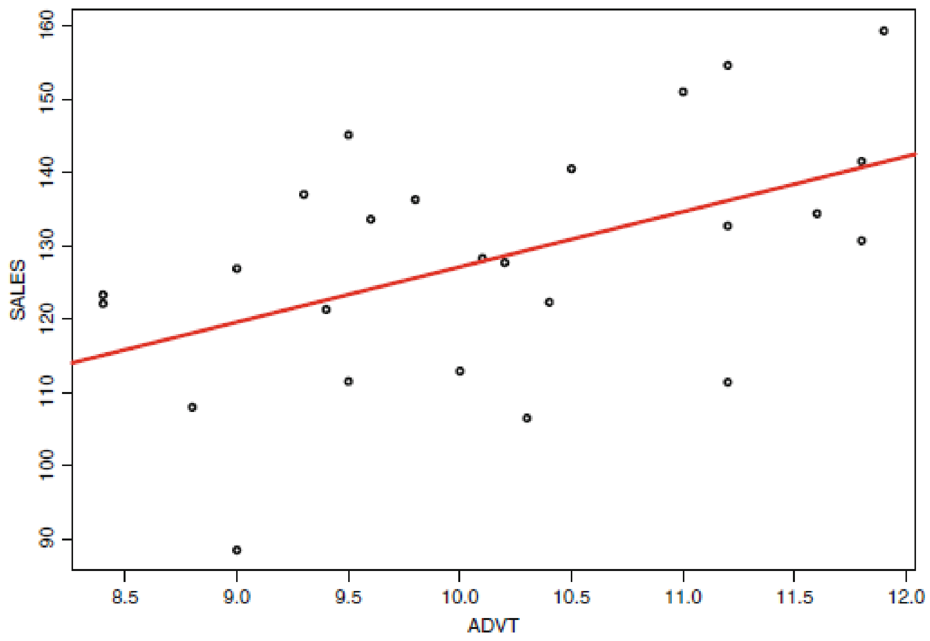
Interpreting Regression Model

- **Interpreting the intercept a = 51.849:** shows that the estimate for the coefficient a equals 51.849, the intercept of the regression line.
- What does this value mean?
  - It means that 51.849 equals the amount of sales with zero advertising.

- **Interpreting the Slope *b* = 7.527:** shows that the estimate for the coefficient *b* equals 7.527, the *slope* of the regression.
  - *b* = 7.527 implies that for every $1*increase* in advertising sales increase by $7.527

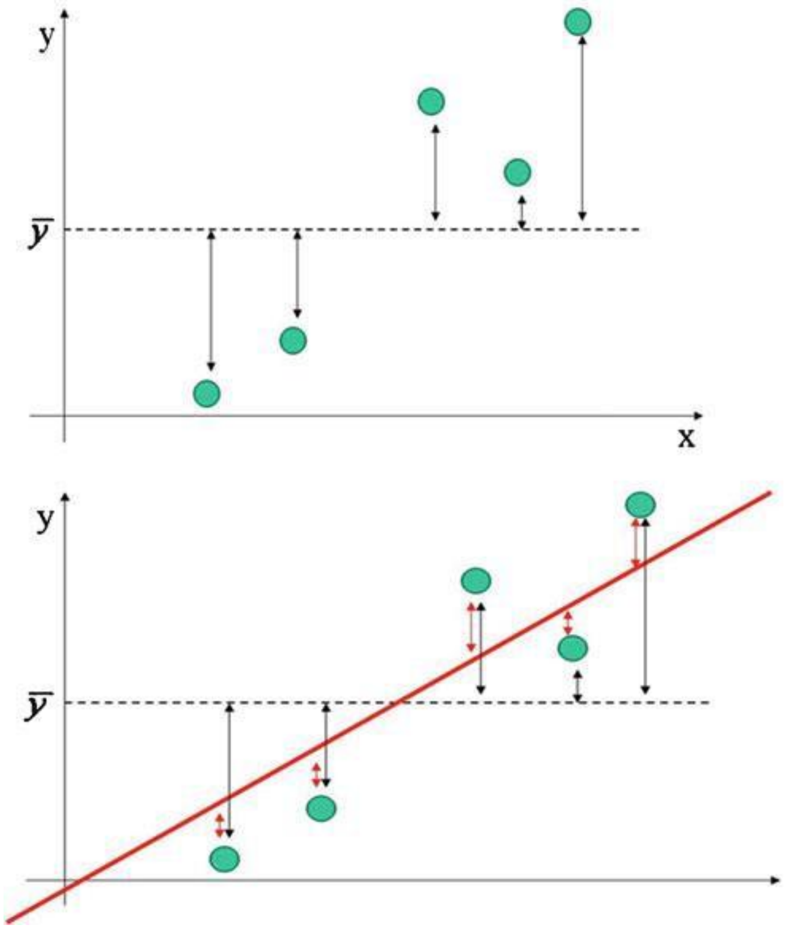
$$\text{Sales} = 51.849 + 7.527 \times \text{Advertising}$$



Evaluating a Regression Model

- **Multiple R-squared, "R-squared" or R^2:** measures the overall quality of a regression model.
  - By "quality" we mean a measure of how well the (idealized) model tracks the (actual) data.

- How good is this model?
  - One way to assess the quality of that model is via its **variance**:
  - the arrows between the data points and dotted line measure the deviation of each individual sales region and the overall model;
  - Larger deviations indicate sales regions where the model provides a poorer fit.



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Total sum of squares (SST)

- Squaring all deviations and summing the squared deviations leads to the concept of the **total sum of squares (SST):**

$$SST = \sum_{i=1}^{n} (y_i - \bar{y})^2$$

- SST is related to the concept of the *sample variance* and measures the *overall variability in the data.*

- What is **variability**?

  -Variability is often referred to as *uncertainty*: the more variability there is among sales regions, the higher our uncertainty about the exact performance of any one sales region in particular.

- The higher the variability, the harder our modeling problem. We take SST as a *benchmark* for our modeling efforts.

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Error sum of squares (SSE)

- How much of the total uncertainty (SST) has the regression line managed to “model away”?
  - There is still some uncertainty left, even after applying the regression model.

- How much is left?
  - We can sum all squared deviations, which leads to the concept of the error sum of squares (SSE).

- **SSE measures** how much variability (or uncertainty) there is left in the data after applying the regression model.

$$SSE = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

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Sum of squares Regression (SSR)

- **SSR** is a measure for how much the model has helped reduce the uncertainty by combining the concepts of *overall uncertainty (SST)* and *uncertainty left over after applying* the regression model (*SSE*) :

$$\mathrm{SSR} = \mathrm{SST} - \mathrm{SSE}$$

- **SSR** quantifies the uncertainty that the regression model was able to model away.

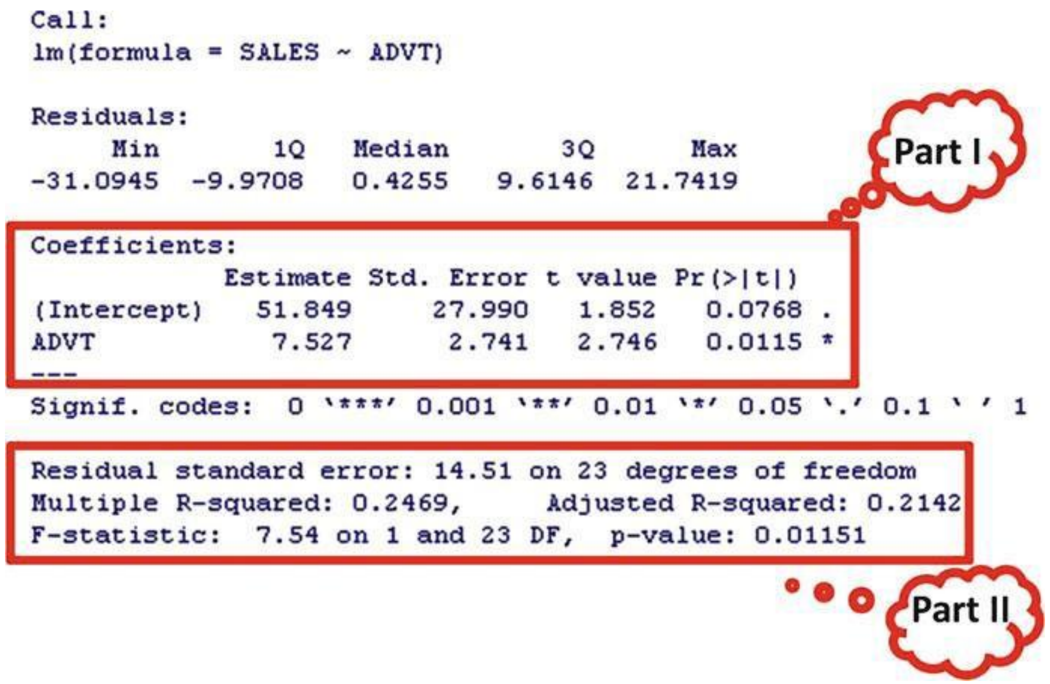
- The proportion of the total variability that the regression was able to capture is **R-squared**:

$$R\text{-}squared = \frac{SSR}{SST}$$

Interpreting R-squared

- **Interpreting R-squared = 0.2469:** our regression model explains 24.69% of the total variability in sales.
  - Is this good or bad?

- The perceived quality of R-squared depends on the context.
  - For Chemical lab Experiment would an R-squared value of 24.69% as outrageously poor.
  - For social scientists a 24.69% R-squared is reasonably high



```
Call:
lm(formula = SALES ~ ADVT)

Residuals:
     Min       1Q   Median       3Q      Max
-31.0945  -9.9708   0.4255   9.6146  21.7419

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   51.849     27.990   1.852   0.0768 .
ADVT           7.527      2.741   2.746   0.0115 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 14.51 on 23 degrees of freedom
Multiple R-squared: 0.2469,     Adjusted R-squared: 0.2142
F-statistic:  7.54 on 1 and 23 DF,  p-value: 0.01151
```

Comparing Regression Models

Another regression model:

- The only difference compared with the previous model is a second explanatory variable, “INCOME,” which denotes the median household income in each sales region.

```
Call:
lm(formula = SALES ~ ADVT + INCOME)

Residuals:
     Min       1Q   Median       3Q      Max
-22.6876  -6.0687   0.6043   7.0523  28.5334

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  36.8948    24.9629   1.478   0.1536
ADVT          5.0691     2.5397   1.996   0.0585 .
INCOME        0.8081     0.2816   2.870   0.0089 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 12.66 on 22 degrees of freedom
Multiple R-squared: 0.452,     Adjusted R-squared: 0.4022
F-statistic: 9.074 on 2 and 22 DF,  p-value: 0.001338
```

- The formal model is now:

$$\text{Sales} = a + b_1 \times \text{Advertising} + b_2 \times \text{Income}$$

- Which model is better?

Comparing R-squared Values

- The **R-squared value** for the first model equals 24.69%, and that for the model in second model equals 45.20%
  - clearly, the second model explains a larger proportion of the total uncertainty in sales and is thus a better model.
- **R-squared** has the unpleasant property that it will **never decrease**, even if we add variables to the model that are complete nonsense.
  - Thus, we should not overly rely on R-squared alone when comparing models
- **Adjusted R-squared** penalizes the model for the inclusion of nonsense variables, and hence we can use it to compare models with different variables.

Testing for Statistical Significance

- **p-value** measures the probability that, given the current information, a **particular predictor** has **no relationship** with the response.
- The **lower the p-value**, the **higher the statistical importance** of this predictor.
- The **smaller this probability**, the more confident we are that the **observed signal is "real" and not** occurred **simply due to random chance**.
- Typically, a variable is considered as **insignificant** (or statistically unimportant) if the associated **p-value is larger than 0.05**.

Gauging Practical Importance

- a predictor is statistically important (i.e., its signal is large relative to its noise) does not imply that it is also **practically useful.**
  - In other words, while statistical importance is a necessary condition for practical importance, it is not sufficient.
- **For example**; deciding whether or not to add another bathroom to our house.
- Only makes sense economically if it adds at least $6,000 to the house is value. Should we add that bathroom?
  - The slope of the variable bathrooms equals 7,883.278; implies that, for each additional bathroom, on average the price of the house increases by 7,883.
- How can we quantify this uncertainty and come to a statistically sound decision?

```
Coefficients:
                     Estimate Std. Error t value Pr(>|t|)
(Intercept)           598.919   9552.197   0.063 0.950110
SqFt                   52.994      5.734   9.242 1.10e-15 ***
Bedrooms             4246.794   1597.911   2.658 0.008939 **
Bathrooms            7883.278   2117.035   3.724 0.000300 ***
Offers              -8267.488   1084.777  -7.621 6.47e-12 ***
BrickYes            17297.350   1981.616   8.729 1.78e-14 ***
NeighborhoodNorth    1560.579   2396.765   0.651 0.516215
NeighborhoodWest    22241.616   2531.758   8.785 1.32e-14 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10020 on 120 degrees of freedom
Multiple R-squared: 0.8686,     Adjusted R-squared: 0.861
F-statistic: 113.3 on 7 and 120 DF,  p-value: < 2.2e-16
```

Confidence interval

- The **concept of confidence intervals**: we compute a 95% confidence interval for the slope of bathrooms as

  slope of bathrooms±(2)*(standard error)

  or, filling in the values from; $7,883.278±(2)*($2,117.035) , which equals *($3,649.208, $12,117.35)*

- The smallest possible added premium due to an additional bathroom is as low as $3,649.208, which is much lower than our desired value of $6,000;
  - hence we should not add the bathroom to our house.

Flexible regression models

Limitation of Linearity

- Linearity is a basic feature of regression
- It is efficient to implement and easy to interpret.
- but also one of its main limitations.
- Linearity is often only an approximation to reality.
  - Many phenomena and processes that occur in nature or in business are not linear at all.
- Using a linear regression model, we often trade **convenience** for **accuracy**.

For Example

- A regression model may tell you that the more you put into advertising, the higher your sales will be.
- While more advertising generally results in higher sales, consumers may reach a "**saturation point**"
- After that point, additional advertising may not add anything to sales.
- Even be a point after which consumers get annoyed and it may in fact have a negative effect on sales.

Need for flexible models

- Saturation points, diminishing returns, or inverse-U shapes cannot be modeled using linear regression – at least not directly.

- In order to accommodate such effects in model, we need more flexible approaches.

- There exist many ways of making regression more flexible – some rely on mathematical and computationally complex methodology, and others involve rather quick and simple "tricks."

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Interaction Terms and Dummy Variables

- Two important concepts in statistics that help make models more flexible:
  - interaction terms
  - dummy variables

Interaction terms

- An "**interaction term**," on the other hand, is simply the multiplication of two variables.

- While multiplying two variables does not appear to be rocket science, it results in a much more flexible regression model.

- Linear relationship between two variables X and Y implies that, for every increase in X, Y also increases, and it increases at the ***same rate,*** which is very *strict assumption***.**

- Interaction terms allow us to model the rate of increase as a function of a third variable.

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Interaction terms

For example;

- Does consumers' spending increases with their salary regardless of their geographical location?
- Or is it more plausible that consumers' spending increases faster for those who live closer to a relevant store or mall?

Another example:

- Are consumers equally price sensitive, regardless of the channel?
- Or, is it more plausible that price sensitivity is higher for online sales channels?

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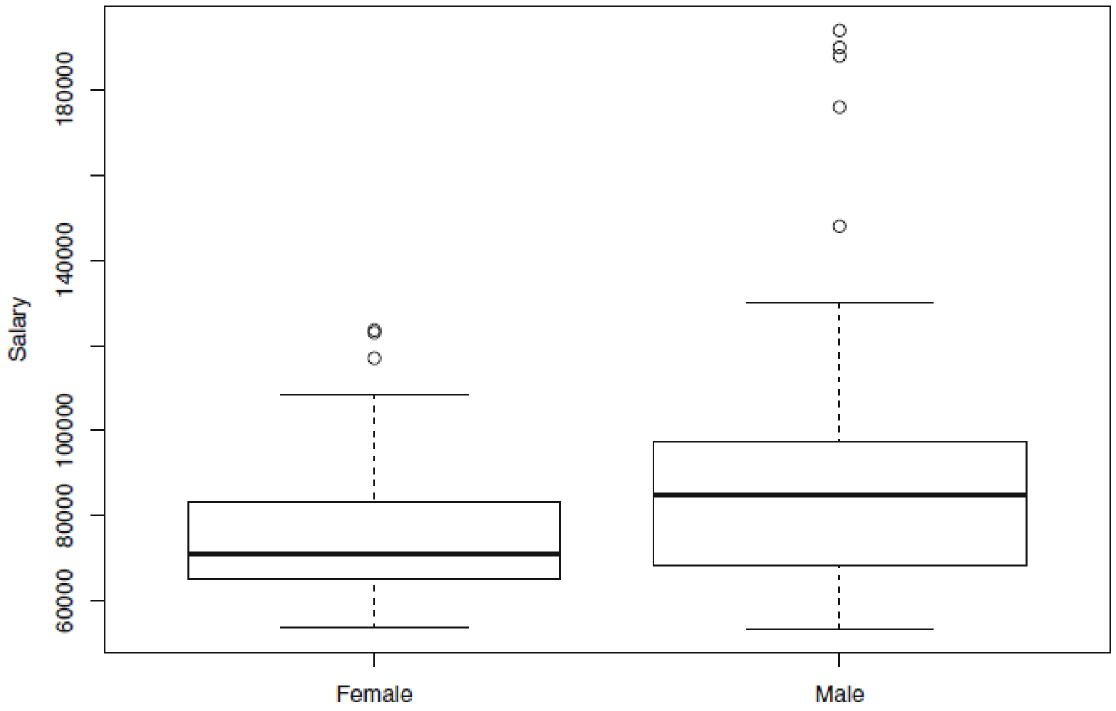
Example

Table shows information about employees' gender, experience and annual salary.

- Is there systematic compensation discrimination against female employees?
- All else equal, do female employees earn less than their male counterparts?

Gender	Experience	Salary
Male	7	53400
Female	11	53600
Female	6	54000
Male	10	54000
Female	5	57000
Female	6	57000
Female	4	57200
Female	11	57520
Male	3	58000

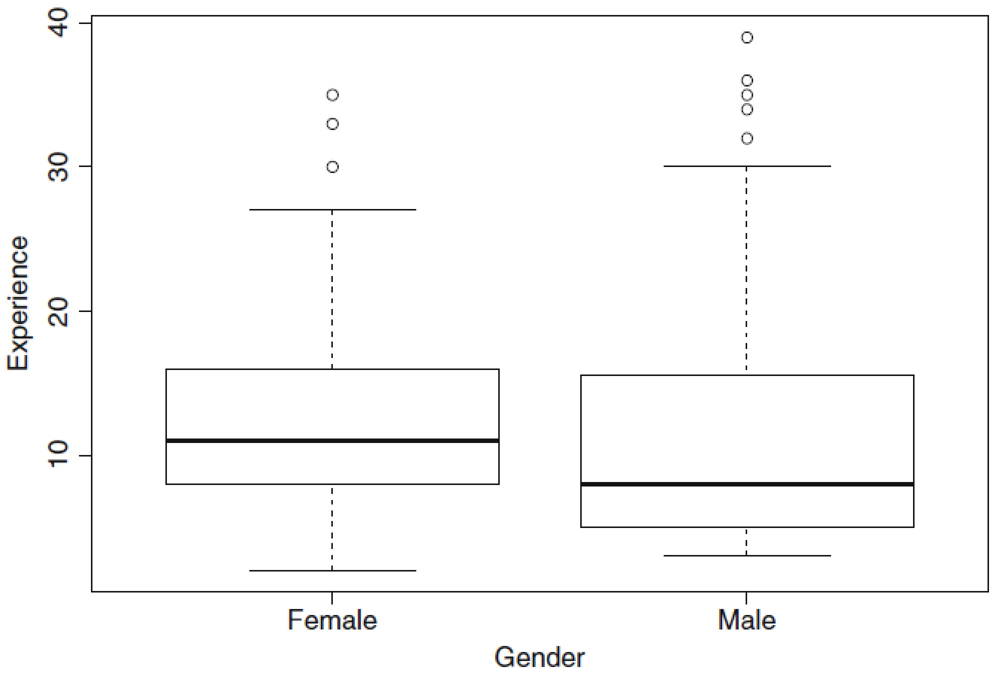


Boxplots of salary, broken up by gender.

- The median salary of female employees is significantly lower than for male employees.
- The highest salary levels for female employees are much lower than those of male employees

- Experience levels of female employees are not lower than those of male employees.
    - The median experience level of female employees is higher than that of their males.

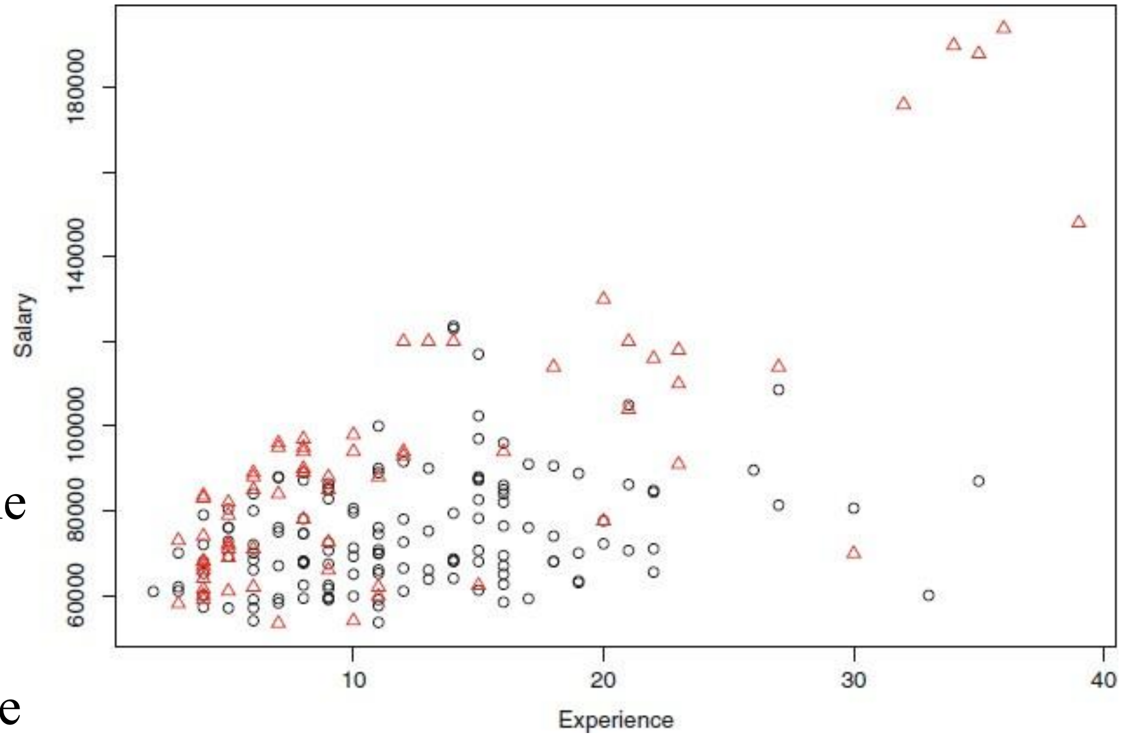
- Low experience levels of female employees can’t be the explanation for their lower salary levels
    - There may be other factors not included in the data causing the salary discrepancy.



Interaction terms

Scatterplot between two pieces of information, salary and experience:

- effect of a third variable (i.e., gender) is also controlled by marking male employees with triangles and female employees with circles.
- salary pattern for male employees (triangles) follows a different path than that of female employees (circles).
- need to derive a model that is flexible enough to accommodate and capture these different salary patterns.



Dummy Variables

- The regression is based on numeric variables as it can only handle numeric data values.
- **“Dummy variable”** is another name for a binary variable, and it simply refers to a recoding of the categoric data.
- However, it is important to note that we do not mean just about any recoding of the data
  - the specific form of the recoding plays an important role

Creating Dummy Variables

- For example; gender

$$\text{Gender.Male} = \begin{cases} 1, & \text{if gender} = \text{“male”} \\ 0, & \text{otherwise} \end{cases}$$

- The new variable Gender.Male only assumes values zero and one, so it is numeric.

- Gender.Male is a **dummy variable** because it does not carry any new information but simply recodes existing information in a numerical way.

- Notice that for a male employee Gender.Male = 1 and for a female employee Gender.Male = 0.
  - Gender.Female carries the identical information as Gender.Male since Gender.Male=1−Gender.Female,

Regression Model with a Dummy Variable

```
Call:
lm(formula = Salary ~ Gender.Male)

Residuals:
   Min     1Q Median     3Q    Max
-37611 -12868  -3720   8230 102989

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)    74420       1789  41.597  < 2e-16 ***
Gender.Male    16591       3129   5.302 2.94e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 21170 on 206 degrees of freedom
Multiple R-squared: 0.1201,	Adjusted R-squared: 0.1158
F-statistic: 28.12 on 1 and 206 DF,  p-value: 2.935e-07
```

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Interpreting a Dummy Variable Regression Model

It shows a regression model with only the gender dummy variable (and salary as the response variable).

- We can see that Gender.Male is statistically significant. (Notice the small p-value.)
- We can also see that its estimated coefficient equals 16,591.

What does this mean?

- Strictly speaking, it implies that for every increase in the dummy variable Gender.Male by one unit, salary increases by $16,591.
  - In other words, male employees make $16,591 more in salary than female employees.

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Insight about Dummy Variables

- Dummy variables always conduct pairwise comparisons.

- When we include a dummy variable in a regression model, the intercept contains the effect of the "baseline."
  - Gender.Male=1defined for male employees; the level that is defined as zero (female in this case) is often referred to as the baseline.
  - The effect of the baseline is contained in the intercept. For example, the intercept denotes the salary level of female employees.

A More Complex Dummy Variable Regression Model

```
Call:
lm(formula = Salary ~ Experience + Gender.Male)

Residuals:
     Min       1Q   Median       3Q      Max
-52779.5  -9806.3   -121.1   8346.8  60912.8

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  53260.0     2416.6  22.039  < 2e-16 ***
Experience    1744.6      160.7  10.858  < 2e-16 ***
Gender.Male  17020.6     2499.6   6.809 1.06e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 16910 on 205 degrees of freedom
Multiple R-squared: 0.4413,     Adjusted R-squared: 0.4359
F-statistic: 80.98 on 2 and 205 DF,  p-value: < 2.2e-16
```

A More Complex Dummy Variable Regression Model

- Estimated regression model;

$$\text{Salary} = 53{,}260 + 1{,}744.6 \times \text{Experience} + 17{,}020.6 \times \text{Gender.Male}$$

How does equation help us characterize female employees?

- Setting the dummy equal to zero, we get the *female-specific* regression equation

$$\text{Salary} = 53{,}260 + 1{,}744.6 \times \text{Experience}$$

- Setting Gender.Male = 1, we get the *male-specific* regression equation

$$\text{Salary} = (53{,}260 + 17{,}020.6) + 1{,}744.6 \times \text{Experience}$$

$$\text{or}\quad \text{Salary} = 70{,}280.6 + 1{,}744.6 \times \text{Experience}$$

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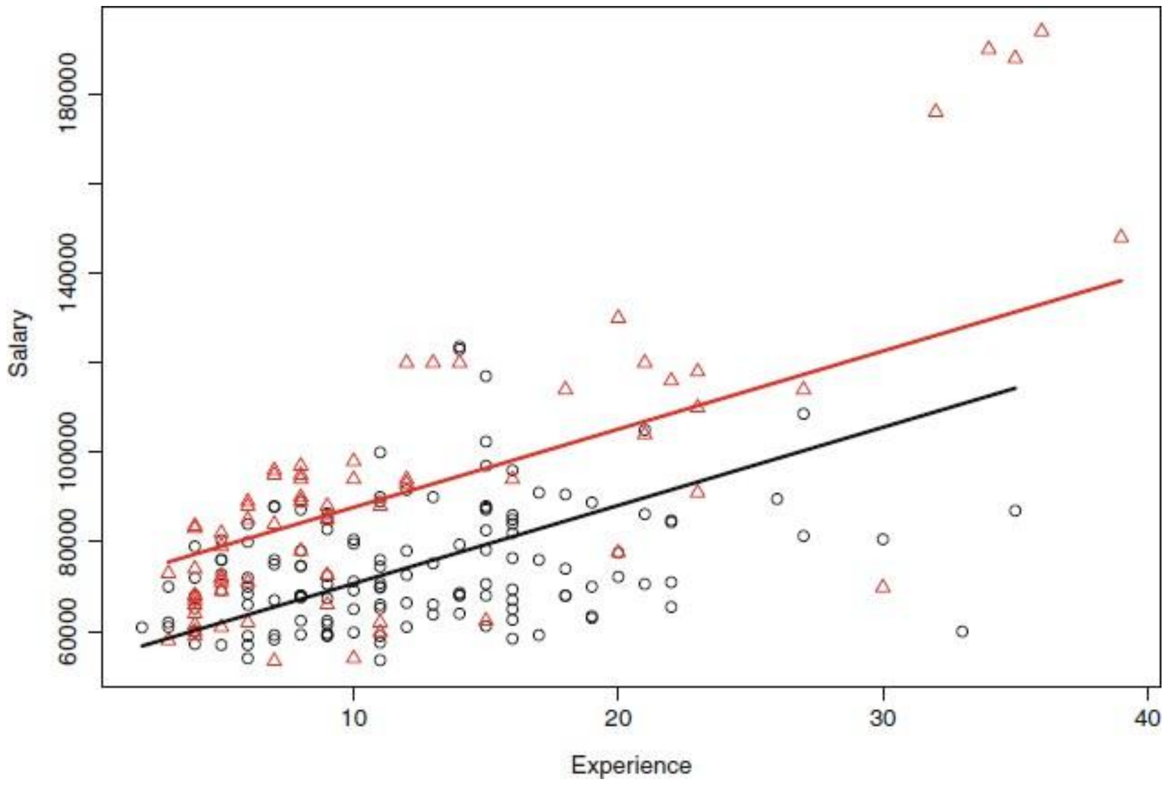
Interpreting More Complex Model with Dummy Variable



- The **impact of experience is the same** for both male and female employees.
  - the incremental impact of an additional year of experience is per our model, identical for male and female employees.

- The only **difference lies in the intercept** (i.e., in the starting salary).
  - per our model, the starting salary is higher for male employees than for their females

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Interaction Terms

- Dummy variable regression models result in increased flexibility and better capture the varying data patterns

- However, the **parallel lines** raise a question.
  - doesn't it appear as if for every additional year of experience salary for male employees grows at a faster rate than that of female employees?

- Non-parallel lines imply slopes that vary – but how can we introduce varying slopes into our regression model?
  - The answer can be found in the concept of **interaction terms.**

Creating Interaction Terms

- An interaction term is simply the multiplication of two variables.
- For example, multiplying the dummy variable Gender.Male with Experience;

$$\text{Gender.Exp.Int} = \text{Gender.Male} \times \text{Experience}$$

- The (row-by-row) multiplication of Gender.Male and Experience gives

$$\text{Gender.Male} \times \text{Experience} = \begin{pmatrix} 1 \times 7 \\ 0 \times 11 \\ 0 \times 6 \end{pmatrix}$$

or

$$\text{Gender.Exp.Int} = \begin{pmatrix} 7 \\ 0 \\ 0 \end{pmatrix}$$

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Interaction Terms Model

```
Call:
lm(formula = Salary ~ Experience + Gender.Male + Gender.Exp.Int

Residuals:
   Min     1Q Median     3Q    Max
-71048  -9278  -1701   9166  47932

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)     66333.6     2811.7  23.592  < 2e-16 ***
Experience        666.7      206.5   3.228  0.00145 **
Gender.Male     -8034.3     4110.6  -1.955  0.05201 .
Gender.Exp.Int   2086.2      287.3   7.261 7.95e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 15110 on 204 degrees of freedom
Multiple R-squared: 0.5561,     Adjusted R-squared: 0.5495
F-statistic: 85.18 on 3 and 204 DF,  p-value: < 2.2e-16
```

Interpreting Interaction Terms

The associated regression equation;

$$\text{Salary}=66{,}333.6+666.7\times\text{Experience}-8{,}034.3\times\text{Gender.Male}+2{,}086.2\times\text{Gender.Exp.Int}$$

- This model provides a better fit to the data, as the value of R-squared is now significantly higher.
  - R-squared equals only 12.01% in first model, it increases to 44.13% in second model. The model with the interaction term provides an even better fit, as the R-squared value 55.61%.
- The p-value of the interaction term is very low, the p-value of the dummy variable is rather large and hence Gender.Male is only borderline significant.
- Also the sign of the dummy variable has changed: it was positive before, it is negative for the model including the interaction term.

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Interaction Terms Regression Model

- Estimated regression model;

  $\text{Salary}=66{,}333.6+666.7\times\text{Experience}-8{,}034.3\times\text{Gender.Male}+2{,}086.2\times\text{Gender.Exp.Int}$

How does equation help us characterize female employees?

- Setting the dummy equal to zero, we get the *female-specific* regression equation

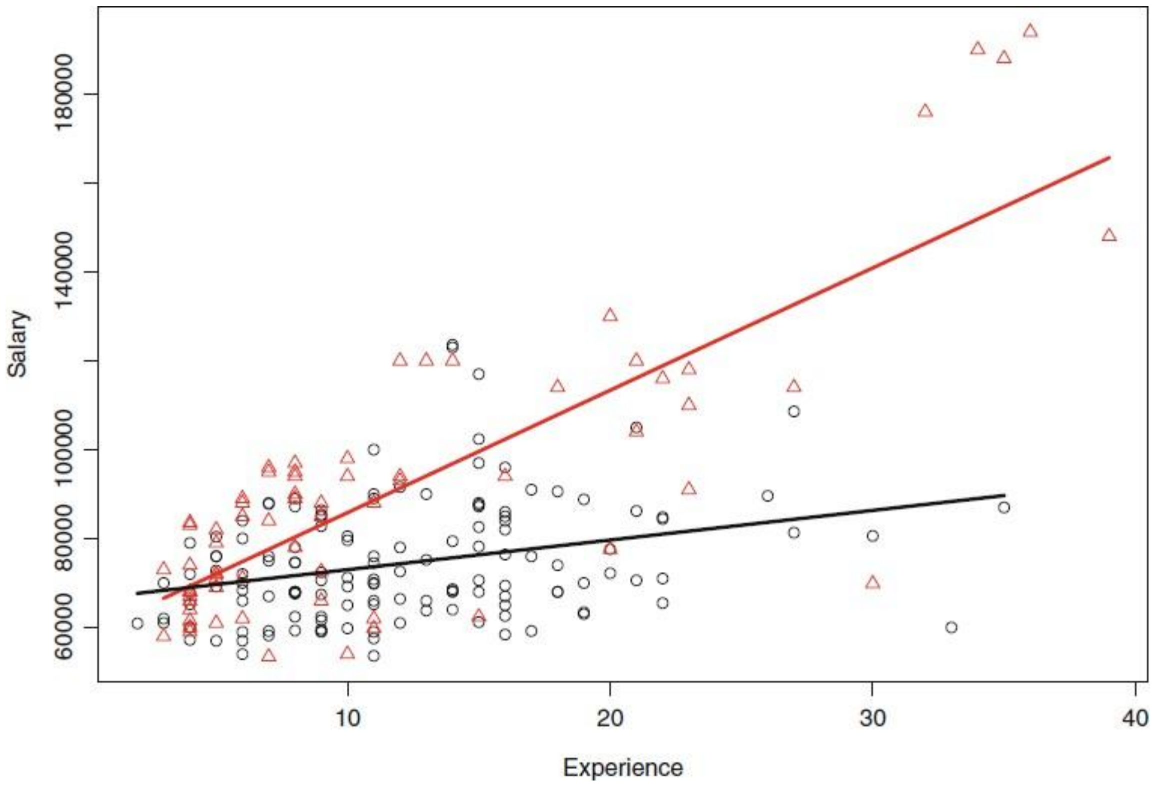
$$\text{Salary} = 66{,}333.6+666.7\times\text{Experience}$$

- Setting Gender.Male = 1, we get the *male-specific* regression equation

  $\text{Salary}=66{,}333.6+666.7\times\text{Experience}-8{,}034.3\times 1+2{,}086.2\times 1\times\text{Experience}$

  or $\text{Salary} = 58{,}299.3+2{,}752.9\times\text{Experience}$

Interpreting Model with Interaction Terms



- The intercept in the male-specific model is lower than that of female employees.
  - starting salary for males is lower than for female employees.
- The slope for experience in the male-specific model is much larger than for female employees.
  - the rate of salary increase for male employees is higher for each additional year of experience.

SDU

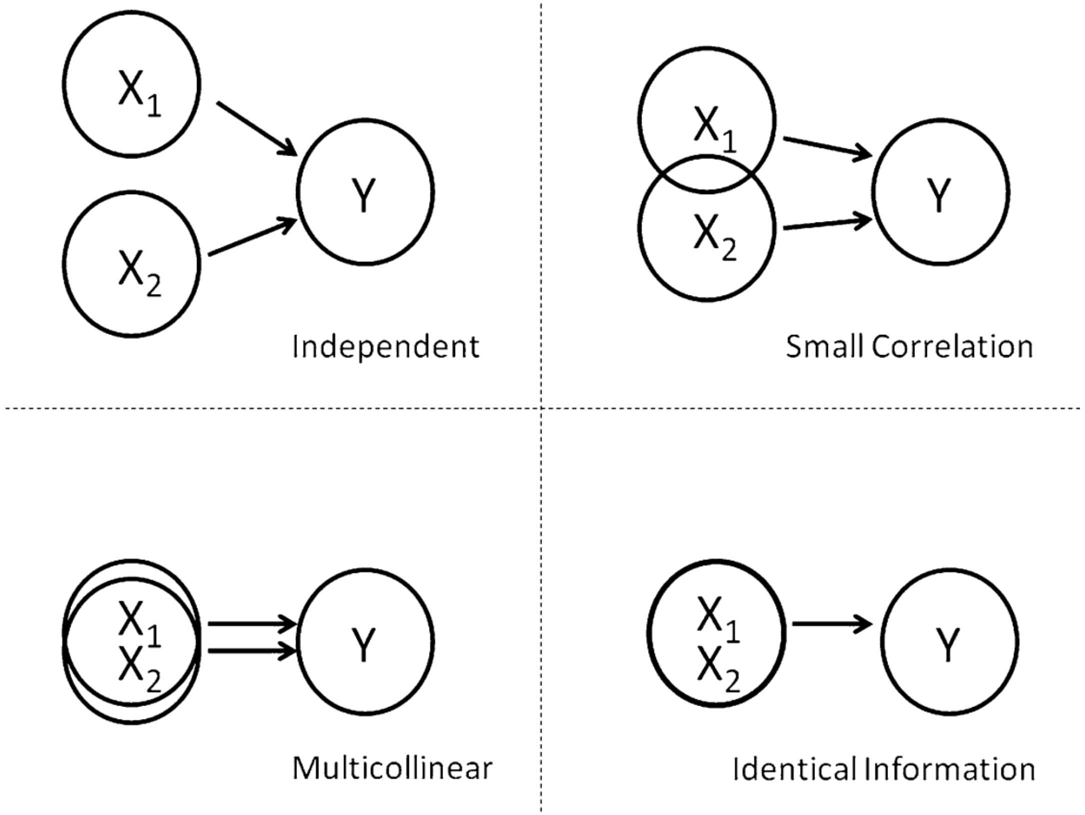
Models with Too Many Predictors: Multicollinearity and Variable Selection

- While the availability of information (i.e., data) is necessary for modeling, it is important to focus **only on the relevant information.**

- Incorporating **too much information** into a regression model can render the model useless.

- Unfortunately, with the availability of terabytes of data in hundreds of data warehouses, **filtering out** the relevant information is as hard as finding the famous needle in a haystack.

- **Multicollinearity** refers to the situation where two (or more) predictors are (strongly) correlated.



Diagnosing Multicollinearity

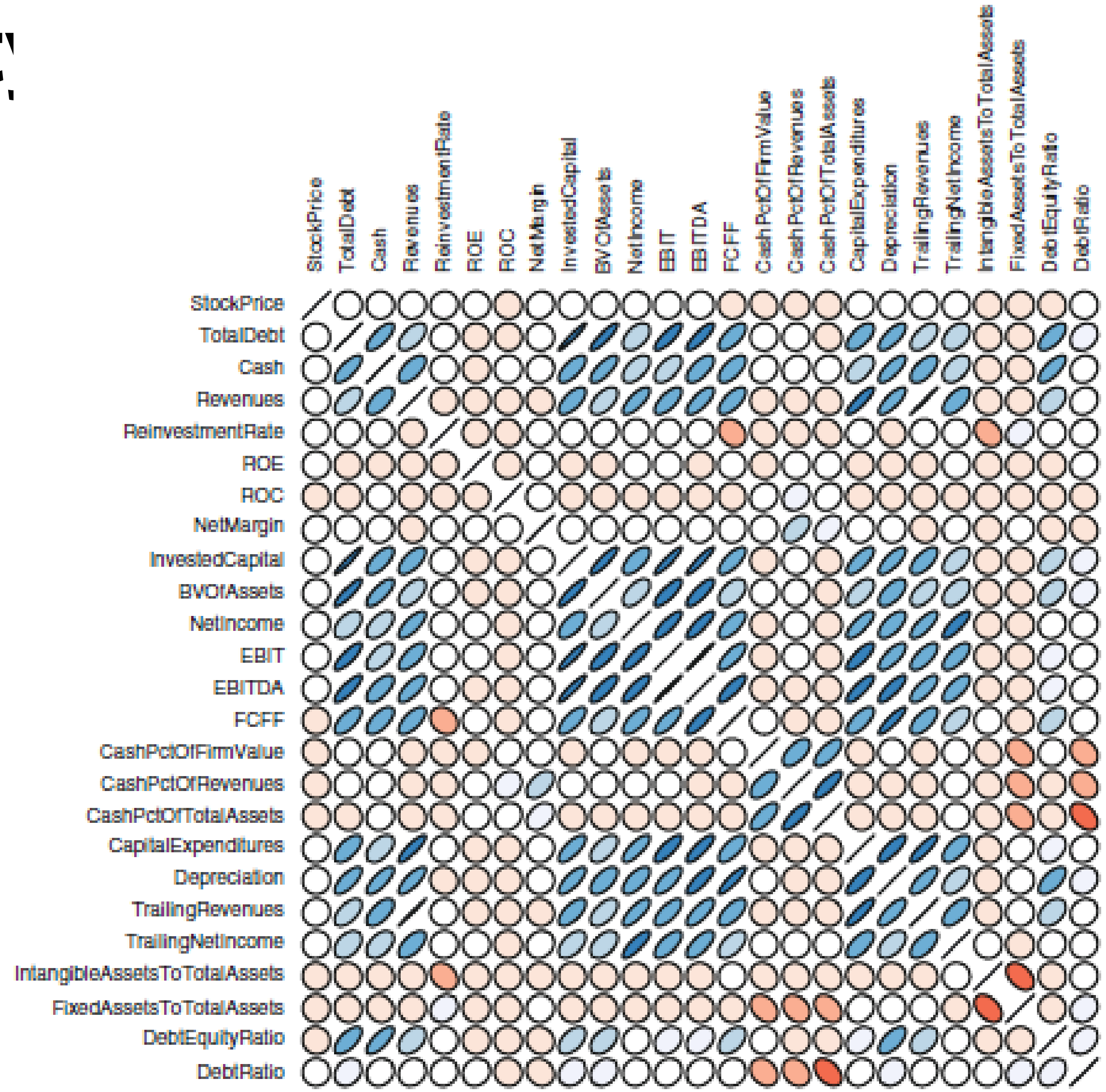
- Most common tools for detecting this problem are **correlation tables** and **scatterplots**.

- The stronger the correlation between X1 and X2, the more we should worry. A correlation of 0.8 or 0.9 should always raise concern

	Sales	Assets
Sales	1	0.9488454
Assets	0.9488454	1

- The most extreme situation (i.e., a correlation equal to one) , where we code the identical piece of information in two different ways such as units in Kg and Grams.

- The cure for multicollinearity is, at least conceptually, very simple, and it is given by the very careful selection of variables.



Thanks for your attention